SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 04.2

Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000;        % Passband cutoff (Hz)
fs = 4000;        % Sampling frequency (Hz)
N  = 50;          % Filter order (even number preferred)
wc = fp/(fs/2);   % Normalized cutoff frequency (0 to 1)
%% --- FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- Plot Frequency Response ---
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

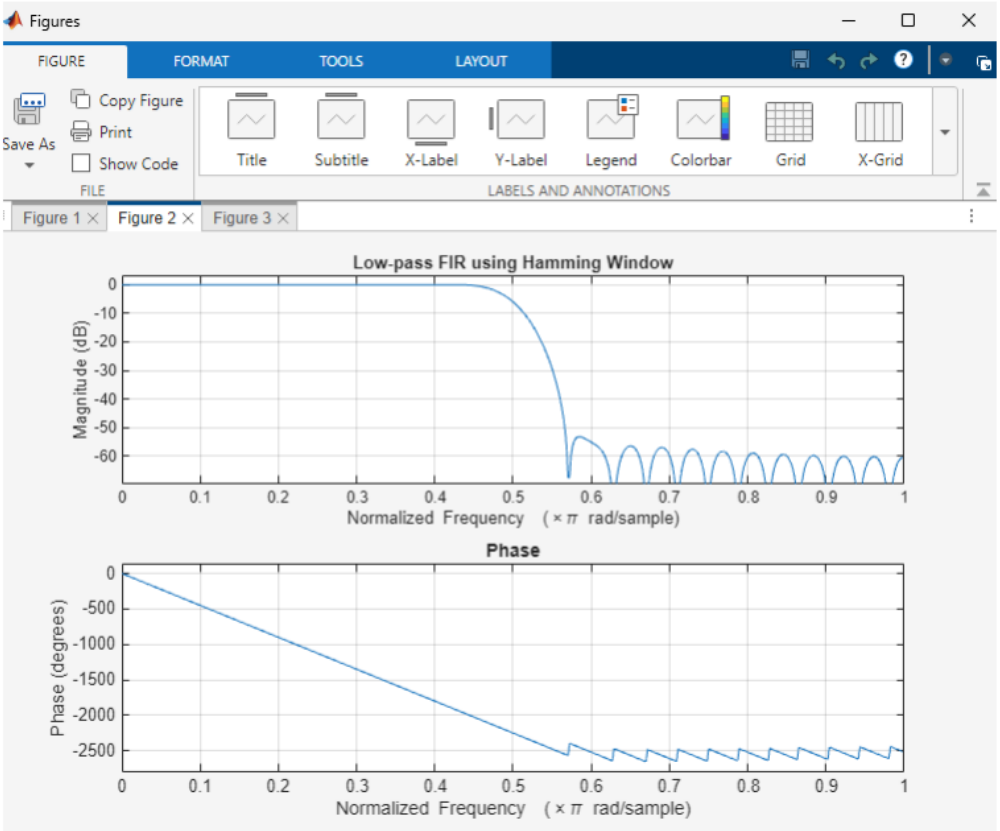
```
b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm');

%% --- Plot Frequency Responses ---

figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');
```

Output

```
--- Hamming Window Coefficients ---
0.0010
-0.0000
-0.0013
0.0000
0.0021
-0.0000
-0.0034
0.0000
0.0055
-0.0000
-0.0084
0.0000
0.0125
-0.0000
-0.0181
0.0000
0.0260
-0.0000
-0.0379
0.0000
0.0580
-0.0000
-0.1026
0.0000
0.3168
0.4995
0.3168
0.0000
-0.1026
-0.0000
0.0580
```

0.0000
-0.0379
-0.0000
0.0260
0.0000
-0.0181
-0.0000
0.0125
0.0000
-0.0084
-0.0000
0.0055
0.0000
-0.0034
-0.0000
0.0021
0.0000
-0.0013
-0.0000

0.0010



Result

1. The FIR Low-Pass Filter was successfully designed using the Hamming Window technique for the specified filter parameters.
2. The FIR filter coefficients were generated and implemented successfully in MATLAB.
3. The magnitude and phase responses of the designed filter were obtained and analyzed.
4. The filter exhibited the desired low-pass characteristics with improved stopband attenuation compared to the Rectangular Window method.

Practical Question 1

Design an FIR Low-Pass Filter using the **Hamming Window** technique in MATLAB for a sampling frequency of **8000 Hz**, cutoff frequency of **1500 Hz**, and filter order of **50**.

Tasks:

1. Calculate the normalized cutoff frequency.

2. Generate the FIR filter coefficients using the fir1() function with a Hamming Window.
3. Obtain and plot the magnitude response and phase response of the designed filter.
4. Analyze the passband and stopband characteristics and verify the low-pass filtering behavior.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

Design an FIR Low-Pass Filter using the **Hamming Window** technique in MATLAB for a sampling frequency of **12000 Hz**, cutoff frequency of **2000 Hz**, and filter order of **60**.

Tasks:

1. Design the FIR Low-Pass Filter using the specified parameters and Hamming Window.
2. Generate and display the FIR filter coefficients.
3. Determine the magnitude response and phase response using the freqz() function.
4. Evaluate the filter performance in terms of passband response, stopband attenuation, and transition band characteristics.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To design and analyze an FIR low-pass filter using a Hamming window and evaluate its response.

Student Details

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